

Artificial Intelligence Product Management Case / PRD: Turning Messy User Interviews into Scoped Epics and Prioritized Tickets

1. Why I Chose This Problem

One thing I noticed while working on product-heavy projects is that user research is only useful if it actually turns into execution.

It is pretty easy to say, “We talked to users and found pain points.” The harder part is taking a messy interview transcript, figuring out what the user actually needs, grouping those needs into themes, and turning them into epics, user stories, tickets, and sprint priorities.

This problem feels especially real for student product teams, startup teams, and early-stage PMs. A lot of teams do not have a clean product ops system. They have notes in Google Docs, scattered Slack messages, random interview quotes, and a backlog that gets built mostly from vibes.

So for this case, I designed an AI-assisted product planning tool that helps PMs turn messy user interviews into scoped epics and prioritized tickets.

The goal is not to replace the PM. The goal is to help the PM move faster from research → synthesis → roadmap → execution.

2. Problem Statement

PMs and product teams spend a lot of time manually translating user interviews into actionable product work. This process is slow, inconsistent, and easy to mess up because interview data is usually unstructured.

Main problem:

PMs need a faster and more reliable way to convert messy user feedback into themes, epics, user stories, and prioritized tickets without losing the actual user context behind each decision.

3. Target Users

Primary User: Early-career PM / Student PM / Startup PM

This user is responsible for:

- Running user interviews
- Synthesizing insights
- Writing requirements
- Creating epics and tickets
- Prioritizing what the team should build first

They may not have a dedicated product ops team or research team helping them.

Secondary User: Engineering Lead

This user cares about:

- Clear tickets
- Acceptance criteria
- Reduced ambiguity
- Knowing why something is being prioritized

Secondary User: Designer

This user cares about:

- Understanding user pain points
 - Seeing patterns across interview responses
 - Knowing which flows or wireframes matter most
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4. User Pain Points

From my own experience working on student-led product projects, PM-style projects, and team-based engineering projects, I noticed a few recurring pain points:

1. **Interview notes are messy**
Users do not speak in clean product requirements. They ramble, contradict themselves, and describe problems in emotional or indirect ways.
2. **Synthesis takes forever**
After interviews, the PM still has to identify themes, group pain points, pull quotes, and decide what actually matters.
3. **Backlogs become disconnected from research**
A ticket might say “Add dashboard filters,” but the team may forget which user problem made that feature important in the first place.

4. **Prioritization is often unclear**

Teams say a feature is important, but they do not always explain why. There is not always a clear link between user pain, business impact, and effort.

5. **Junior PMs need structure**

New PMs may know the product problem but not always know how to write strong epics, user stories, or acceptance criteria.

5. Product Goal

Build an AI-powered workspace where a PM can upload messy interview notes or transcripts and get:

- Key user pain points
- Research themes
- Supporting quotes
- Suggested epics
- User stories
- Acceptance criteria
- Prioritized tickets
- A traceable link between user feedback and product decisions

The product should make PMs faster, but still leave final judgment to the PM.

6. Non-Goals

This tool is not trying to:

- Fully automate product strategy
- Replace user researchers
- Replace PM judgment
- Make final roadmap decisions without human review
- Generate generic Jira tickets with no context

The AI should act more like a product thinking assistant, not a product manager.

7. Example User Flow

Step 1: PM uploads research notes

The PM uploads:

- Interview transcript
- Google Doc notes
- Survey responses
- Call notes
- Raw bullet points

Step 2: AI extracts pain points

The tool identifies repeated problems, emotional friction points, user goals, and blockers.

Example output:

Users feel overwhelmed when they have too many club applications open at once and no central place to track deadlines.

Step 3: AI groups insights into themes

Example themes:

- Discovery friction
- Application tracking confusion
- Deadline anxiety
- Lack of transparency after applying

Step 4: PM reviews and edits themes

The PM can approve, delete, merge, or rename themes.

Step 5: AI suggests epics

Example epics:

- Club Discovery Experience
- Application Tracking Dashboard
- Deadline Reminder System
- Club Profile Standardization

Step 6: AI generates user stories and acceptance criteria

Example:

User Story:

As a student, I want to save clubs I am interested in so that I can come back to them before the application deadline.

Acceptance Criteria:

- Student can save a club from the club profile page
- Saved clubs appear in “My Applications”
- Student can remove a saved club
- Saved status persists after refresh

Step 7: AI recommends prioritization

The tool scores tickets based on:

- Frequency of pain point
- Severity of user problem
- Business/product impact
- Engineering effort
- Confidence level from research

Step 8: PM exports to execution tools

The PM can export to:

- Jira
- Linear
- Trello
- Notion
- Google Sheets

8. Core Features

Feature 1: Interview Upload + Parsing**Description:**

Allow PMs to upload raw user interview notes, transcripts, or survey responses.

Why it matters:

This is the starting point of the workflow. The tool has to accept messy input because real research is rarely clean.

Requirements:

- Support copy-paste text
 - Support Google Doc-style notes
 - Support transcript uploads
 - Detect speaker sections when available
 - Highlight unclear or low-confidence sections
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Feature 2: AI Theme Extraction

Description:

Automatically group user feedback into themes and pain points.

Why it matters:

This saves PMs time and helps them see patterns faster.

Requirements:

- Extract repeated user pain points
 - Group similar feedback into themes
 - Show supporting quotes
 - Display confidence level for each theme
 - Allow PM to edit, merge, or delete themes
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Feature 3: Epic Generator

Description:

Turn approved themes into scoped product epics.

Why it matters:

A common gap in product work is moving from “here is what users said” to “here is what we should build.”

Requirements:

- Generate epic title
 - Generate problem statement
 - Include linked user pain points
 - Include affected persona
 - Include suggested success metric
 - Allow manual editing
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Feature 4: User Story + Acceptance Criteria Generator

Description:

Generate clear user stories and acceptance criteria for each epic.

Why it matters:

This helps engineering teams get sprint-ready tickets instead of vague product ideas.

Requirements:

- Generate user stories in “As a user, I want...” format
 - Generate acceptance criteria
 - Include edge cases
 - Include dependencies
 - Include open questions for PM review
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Feature 5: Prioritization Assistant

Description:

Score generated tickets using a lightweight prioritization model.

Why it matters:

The PM still decides what gets built, but the tool gives a starting point that is grounded in research.

Suggested scoring inputs:

- User pain severity
- Number of users affected
- Product impact
- Engineering effort
- Confidence level

Example score model:

Priority Score =

$(\text{User Pain} \times \text{Frequency} \times \text{Product Impact} \times \text{Confidence}) / \text{Engineering Effort}$

Feature 6: Research-to-Ticket Traceability

Description:

Each ticket should show which user quotes or themes led to it.

Why it matters:

This prevents teams from losing the “why” behind a feature.

Requirements:

- Each ticket links back to source quotes
 - Each epic links back to research themes
 - PM can view supporting evidence before approving a ticket
 - Export should preserve the source context
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9. MVP Scope

For the MVP, I would keep the product focused on one clean workflow:

Upload interview notes → Extract themes → Generate epics → Generate tickets → Prioritize → Export

MVP Features

1. Text upload or paste interview notes
2. Theme extraction with supporting quotes
3. Epic generation
4. User story + acceptance criteria generation
5. Basic prioritization scoring
6. Export to CSV / Notion-style table

Not in MVP

- Live call transcription
- Full Jira integration
- Team collaboration
- Advanced analytics
- Custom AI model training
- Multi-language support

I would intentionally keep the MVP simple because the main risk is not whether we can add integrations. The main risk is whether PMs trust the AI-generated synthesis enough to use it.

10. Prioritization

I would prioritize based on user value and risk reduction.

Priority	Feature	Reason
P0	Upload/paste interview notes	Needed for the core workflow
P0	Theme extraction	Main value prop of the product
P0	Supporting quotes	Builds trust in the AI output
P0	Epic generation	Connects research to planning
P1	User stories + acceptance criteria	Makes output execution-ready
P1	Prioritization scoring	Helps PM decide what comes first
P2	Export to Jira/Linear	Useful, but not needed to validate MVP
P2	Collaboration features	Better for later-stage teams

11. Success Metrics

Activation Metrics

- % of users who upload their first interview notes
- % of users who generate at least one set of themes
- % of users who approve or edit AI-generated themes

Engagement Metrics

- Number of projects created per user
- Number of interviews processed per project
- Number of generated tickets edited or approved

Quality Metrics

- % of AI-generated themes accepted by PM
- % of generated tickets exported
- Average PM rating of output quality
- Number of manual edits needed per ticket

Business/Product Metrics

- Time saved from interview synthesis to ticket creation

- Retention after first project
- Conversion from free to paid workspace
- Weekly active PMs

North Star Metric

Number of research-backed tickets approved by PMs per week

I like this metric because it captures the actual value of the product. The goal is not just to generate AI text. The goal is to help PMs create usable, research-backed product work.

12. Risks and Tradeoffs

Risk 1: AI output may be too generic

If the generated epics sound like generic PM templates, users will not trust the product.

Mitigation:

Always include source quotes and let users edit/approve themes before ticket generation.

Risk 2: PMs may not want AI making prioritization decisions

PMs may feel like prioritization needs more context than interview notes alone.

Mitigation:

Position prioritization as a recommendation, not an answer. Let PMs adjust scoring inputs.

Risk 3: Bad input creates bad output

Messy notes can be incomplete or unclear.

Mitigation:

Show confidence levels and flag unclear sections.

Risk 4: Privacy concerns

Interview transcripts may include sensitive user or company information.

Mitigation:

Add clear privacy controls, delete options, and workspace-level permissions.

13. Edge Cases

- Interview notes contain multiple users mixed together
 - Transcript has no speaker labels
 - User feedback contradicts itself
 - Only one interview is uploaded, so confidence is low
 - PM uploads notes that are too short
 - AI identifies a theme that is not actually important
 - Generated tickets are too large for one sprint
 - PM wants to merge multiple themes into one epic
 - Team uses a different ticket format
 - Research includes sensitive or private customer data
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14. Launch Plan

Phase 1: Prototype

Build a simple working prototype where a PM can paste notes and get:

- Themes
- Quotes
- Epics
- Tickets

Test with student PMs, hackathon teams, and early-stage builders.

Phase 2: MVP

Add:

- Editable theme review
- Prioritization scoring
- Export to CSV/Notion
- Saved projects

Phase 3: Team Workflow

Add:

- Jira/Linear integrations
- Collaboration
- Comments

- Version history
 - Shared workspaces
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15. Example Output

Raw User Quote

“I honestly don’t know what to do after I interview users. I have pages of notes but turning them into tickets is the part that takes forever.”

Extracted Pain Point

PMs struggle to convert raw interview notes into execution-ready product work.

Theme

Research-to-Execution Gap

Epic

AI-Assisted Research Synthesis

User Story

As a PM, I want interview notes to be grouped into themes so that I can quickly understand the main user pain points.

Acceptance Criteria

- PM can upload or paste interview notes
- AI identifies 3–7 major themes
- Each theme includes supporting quotes
- PM can edit, merge, or delete themes
- Approved themes can be converted into epics

Suggested Priority

P0, because this is the core action that proves the product’s value.

16. My Product Takeaway

The biggest thing I learned from this case is that AI products need to be designed around trust, not just automation.

For this product, the risky part is not generating text. AI can already generate a lot of product-looking content. The harder part is making sure the output is grounded in real user evidence and useful enough for a PM to actually bring into sprint planning.

That is why I would design this product with human review built into every major step. The PM should approve the themes, edit the epics, and own the final prioritization. The AI should help reduce the busywork, but the PM should still own the judgment.